

Unit 2: Computational Thinking and Algorithms

Chapter Importance (Exam Point of View)

- **4 Short Questions** → 8 marks
- **2 MCQs** → 2 marks
- **Total** → 10 marks from this chapter

Focus of the chapter:

- Computational Thinking as a complete process to solve problems
- Algorithms as its most important element
- Later topics will include logical operations (AND, OR, NOT)

2.1 Introduction to Computational Thinking

What is Computational Thinking?

Definition: Computational thinking is a problem-solving approach in which a problem is understood, broken into smaller parts, and solved step by step using logical thinking.

Computational Thinking is a way of thinking to solve problems **clearly and logically**.

Why we use Computational Thinking?

- It helps us break big problems into smaller and simpler parts.
- When small parts are solved, the big problem is automatically solved.

TIDBIT: Computational Thinking is **not limited to computers**. It is used by scientists, engineers, chefs, and in our daily life.

Key Features of Computational Thinking

Computation Thinking means thinking like a problem solver.

1. **Understanding the problem:** Focuses on **understanding the problem** first (step by step).
2. **Decomposition:** We break a large or complex problem into smaller parts and manageable parts.
3. **Pattern Recognition:** We look for **patterns** in problems and solutions.
4. **Abstraction:** We remove unnecessary things and focus only on the main parts.

5. **Design Algorithm:** We create **clear steps** to solve the problem.

These steps are used by **both humans and computers**. (Humans design the algorithm; computers execute it.)

Benefit of Computational Thinking:

Computational thinking helps us think logically and clearly.

Real-Life Examples of Computational Thinking

These concepts are **not limited** to computers or mathematics. Anyone can use them:

1. Cooking a recipe
2. Planning a trip (budget → route → necessary stops)
3. Sorting books in a library (by subject, author, or category)
4. Fixing a mobile phone problem (first find the root cause, then apply steps)

Exam Tips

Short Questions

1. What is computational thinking?
2. Define computational thinking in your own words.
3. Why is computational thinking important?
4. Is computational thinking used only by computer science students? Explain.
5. Name any three key ideas of computational thinking.
6. How does computational thinking help in solving problems?
7. Give two real-life examples where computational thinking is used.
8. What is the main focus of computational thinking?